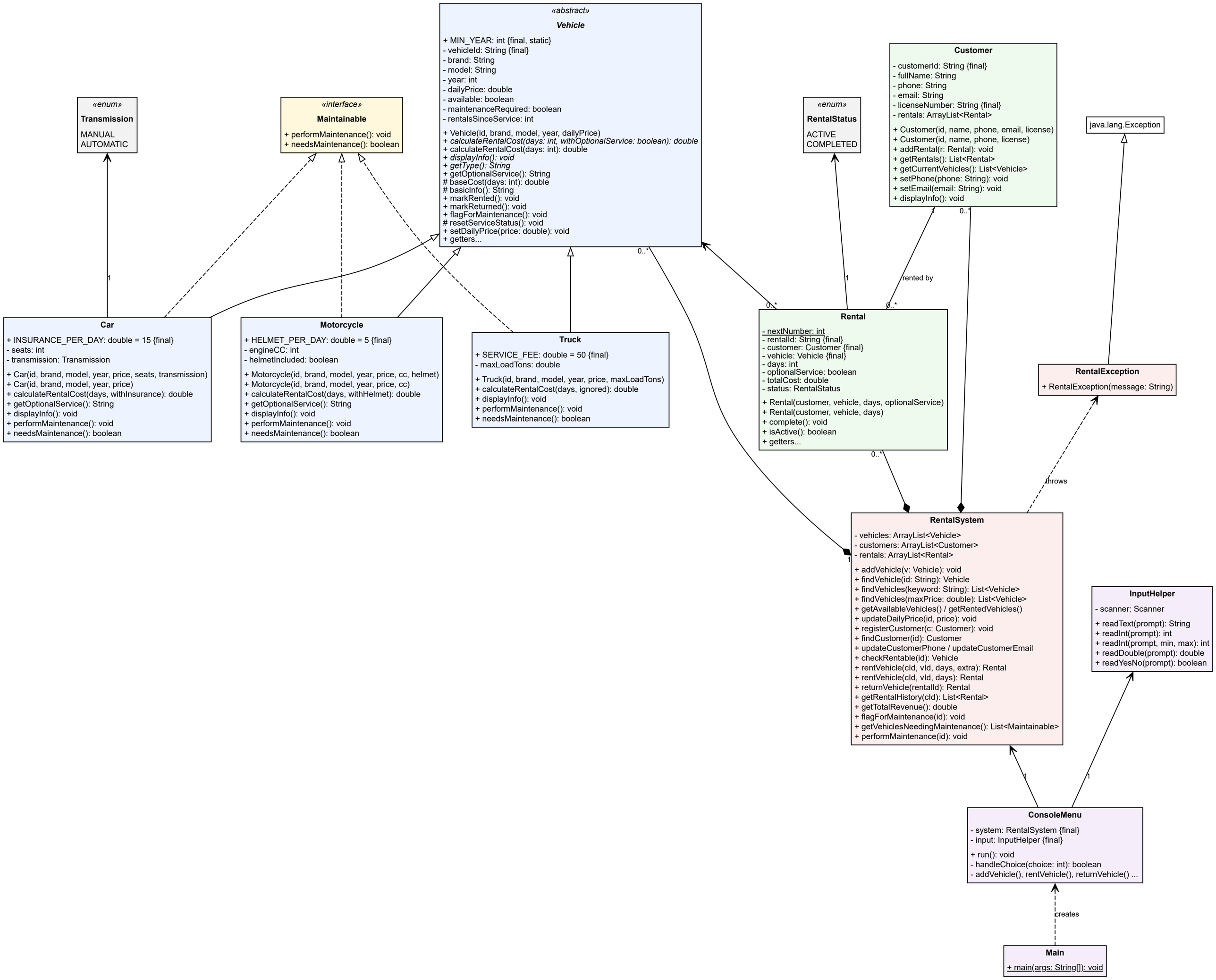

«abstract»
Vehicle
+ MIN_YEAR: int {final, static}
- vehicleId: String {final}
- brand: String
- model: String
- year: int
- dailyPrice: double
- available: boolean
- maintenanceRequired: boolean
- rentalsSinceService: int
+ Vehicle(id, brand, model, year, dailyPrice)
+ calculateRentalCost(days: int, withOptionalService: boolean): double
+ calculateRentalCost(days: int): double
+ displayInfo(): void
+ getType(): String
+ getOptionalService(): String
baseCost(days: int): double
basicInfo(): String
+ markRented(): void
+ markReturned(): void
+ flagForMaintenance(): void
resetServiceStatus(): void
+ setDailyPrice(price: double): void
+ getters...
«enum»
Transmission
MANUAL
AUTOMATIC
«interface»
Maintainable
+ performMaintenance(): void
+ needsMaintenance(): boolean
«enum»
RentalStatus
ACTIVE
COMPLETED
Customer
- customerId: String {final}
- fullName: String
- phone: String
- email: String
- licenseNumber: String {final}
- rentals: ArrayList<Rental>
+ Customer(id, name, phone, email, license)
+ Customer(id, name, phone, license)
+ addRental(r: Rental): void
+ getRentals(): List<Rental>
+ getCurrentVehicles(): List<Vehicle>
+ setPhone(phone: String): void
+ setEmail(email: String): void
+ displayInfo(): void
java.lang.Exception
Car
+ INSURANCE_PER_DAY: double = 15 {final}
- seats: int
- transmission: Transmission
+ Car(id, brand, model, year, price, seats, transmission)
+ Car(id, brand, model, year, price)
+ calculateRentalCost(days, withInsurance): double
+ getOptionalService(): String
+ displayInfo(): void
+ performMaintenance(): void
+ needsMaintenance(): boolean
Motorcycle
+ HELMET_PER_DAY: double = 5 {final}
- engineCC: int
- helmetIncluded: boolean
+ Motorcycle(id, brand, model, year, price, cc, helmet)
+ Motorcycle(id, brand, model, year, price, cc)
+ calculateRentalCost(days, withHelmet): double
+ getOptionalService(): String
+ displayInfo(): void
+ performMaintenance(): void
+ needsMaintenance(): boolean
Truck
+ SERVICE_FEE: double = 50 {final}
- maxLoadTons: double
+ Truck(id, brand, model, year, price, maxLoadTons)
+ calculateRentalCost(days, ignored): double
+ displayInfo(): void
+ performMaintenance(): void
+ needsMaintenance(): boolean
Rental
- nextNumber: int
- rentalId: String {final}
- customer: Customer {final}
- vehicle: Vehicle {final}
- days: int
- optionalService: boolean
- totalCost: double
- status: RentalStatus
+ Rental(customer, vehicle, days, optionalService)
+ Rental(customer, vehicle, days)
+ complete(): void
+ isActive(): boolean
+ getters...
RentalException
+ RentalException(message: String)
RentalSystem
- vehicles: ArrayList<Vehicle>
- customers: ArrayList<Customer>
- rentals: ArrayList<Rental>
+ addVehicle(v: Vehicle): void
+ findVehicle(id: String): Vehicle
+ findVehicles(keyword: String): List<Vehicle>
+ findVehicles(maxPrice: double): List<Vehicle>
+ getAvailableVehicles() / getRentedVehicles()
+ updateDailyPrice(id, price): void
+ registerCustomer(c: Customer): void
+ findCustomer(id): Customer
+ updateCustomerPhone / updateCustomerEmail
+ checkRentable(id): Vehicle
+ rentVehicle(cId, vId, days, extra): Rental
+ rentVehicle(cId, vId, days): Rental
+ returnVehicle(rentalId): Rental
+ getRentalHistory(cId): List<Rental>
+ getTotalRevenue(): double
+ flagForMaintenance(id): void
+ getVehiclesNeedingMaintenance(): List<Maintainable>
+ performMaintenance(id): void
InputHelper
- scanner: Scanner
+ readText(prompt): String
+ readInt(prompt): int
+ readInt(prompt, min, max): int
+ readDouble(prompt): double
+ readYesNo(prompt): boolean
ConsoleMenu
- system: RentalSystem {final}
- input: InputHelper {final}
+ run(): void
- handleChoice(choice: int): boolean
- addVehicle(), rentVehicle(), returnVehicle() ...
Main
+ main(args: String[]): void
rented by
throws
creates
1
0..*